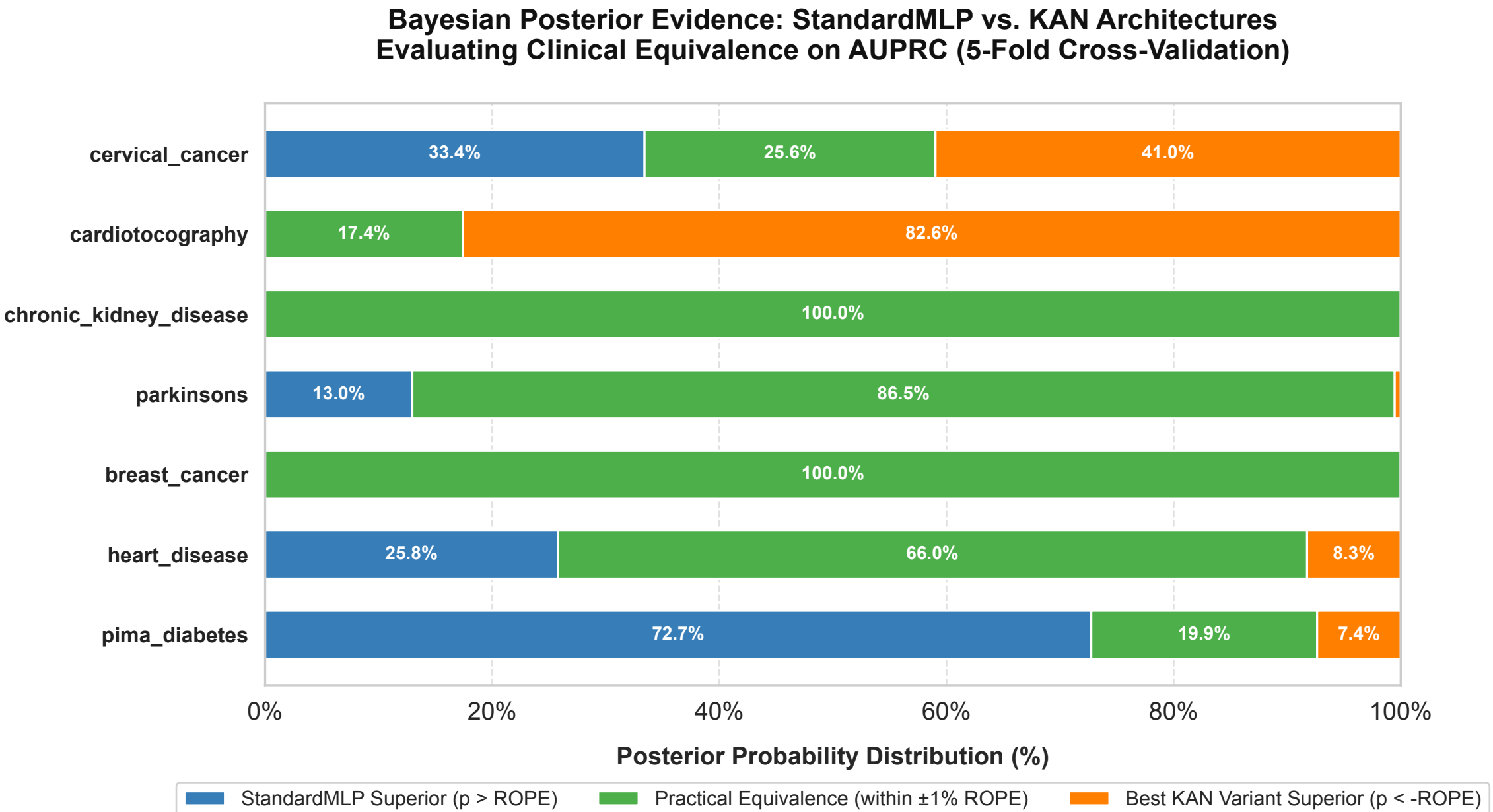
Bayesian Posterior Evidence: StandardMLP vs. KAN Architectures
Evaluating Clinical Equivalence on AUPRC (5-Fold Cross-Validation)
cervical_cancer
cardiotocography
chronic_kidney_disease
parkinsons
breast_cancer
heart_disease
pima_diabetes
33.4%
25.6%
41.0%
17.4%
82.6%
100.0%
13.0%
86.5%
100.0%
25.8%
66.0%
8.3%
72.7%
19.9%
7.4%
0%
20%
40%
60%
80%
100%
Posterior Probability Distribution (%)
StandardMLP Superior (p > ROPE)
Practical Equivalence (within ±1% ROPE)
Best KAN Variant Superior (p < -ROPE)